

DMRG: Density Matrix Renormalisation Group

What it is: Variational principle on the MPS manifold.

$$\text{Minimise } E[\psi] = \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle}$$

over the tensors of an MPS with maximal bond dimension χ .

Freeze everything, except for the tensors on site i and $i+1$ (vary θ , i.e. the two-site α)

$$|\psi\rangle = \sum_{\alpha} \theta_{\alpha} |\varphi_{\alpha}\rangle \quad \alpha = (a, s_i, s_{i+1}, b)$$

$$a - \boxed{\theta} - b$$

$s_i \quad s_{i+1}$

$$|\varphi_{\alpha}\rangle = |\varphi_{a, s_i, s_{i+1}, b}\rangle = |L_a\rangle \otimes |s_i\rangle \otimes |s_{i+1}\rangle \otimes |R_b\rangle$$

$$E = \frac{\langle \psi | H | \psi \rangle}{\langle \psi | \psi \rangle} = \frac{\theta^{\dagger} H_{\text{eff}} \theta}{\theta^{\dagger} G \theta}$$

$$(H_{\text{eff}})_{\alpha\beta} = \langle \varphi_{\alpha} | \hat{H} | \varphi_{\beta} \rangle, \quad G_{\alpha\beta} = \langle \varphi_{\alpha} | \varphi_{\beta} \rangle$$

$$= \delta_{\alpha\beta} \text{ for mixed-canonical.}$$

Important: 1) The energy is a quadratic form in each tensor separately.

So we can optimise one tensor (or OC) at a time, keeping the rest of the MPS fixed.

2) The norm matrix G makes it a generalised eigenvalue problem

$$\frac{\partial E}{\partial \theta_{\alpha}^*} = 0 \Rightarrow \boxed{H_{\text{eff}} \theta = E G \theta}$$

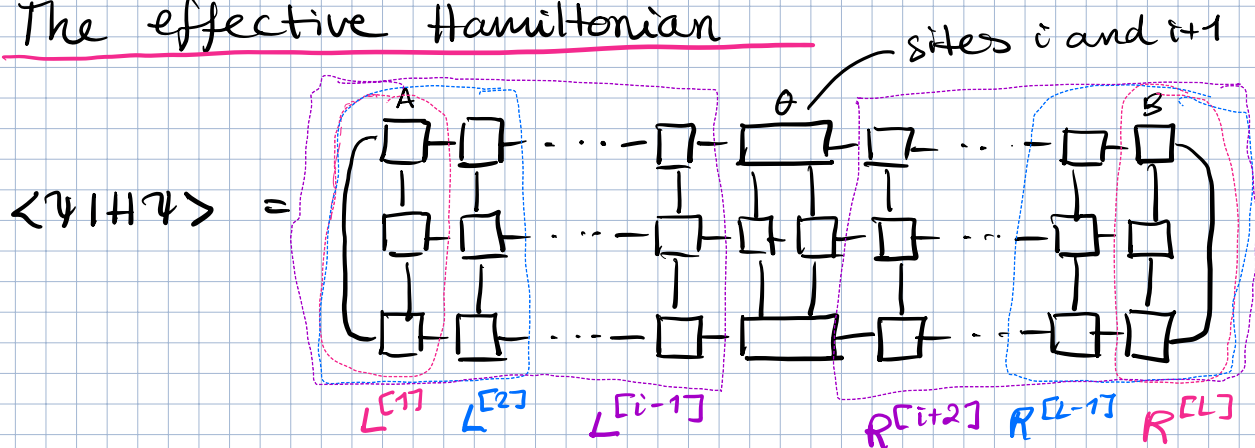
To avoid having to invert G (which would be prone to numerical error), we work with mixed-canonical forms, so that $G = \mathbb{1}$

$$\boxed{H_{\text{eff}} \theta = E \theta}$$

We can now find the optimal C_0 by solving this ordinary eigenproblem. Typically by Lanczos.

Core of DMRG: Sweep across the lattice, optimising θ at sites $(1, 2)$, then move θ to $(2, 3) \dots$ until $(L-1, L)$ and then back.

The effective Hamiltonian



$$= \left[L^{[i-1]} \right] \begin{array}{c} \theta \\ W \\ W \\ \theta \end{array} \left[R^{[i+2]} \right] = \theta^\dagger H_{\text{eff}} \theta$$

$$H_{\text{eff}} = \left[L^{[i-1]} \right] \begin{array}{c} W_i \\ W_{i+1} \end{array} \left[R^{[i+2]} \right]$$

* 4 physical indices
 * 4 bond indices

$\dim H_{\text{eff}} = (x^2 d^2) \times (x^2 d^2)$
 (Compared to $\dim H = d^L \times d^L$)

Object	Dimension
Θ	$\chi^2 d^2$ - vector (χ, d, d, χ)
H_{eff}	$(\chi^2 d^2) \times (\chi^2 d^2)$ - matrix $(\chi, \chi, d, d, d, d, \chi, \chi)$
L/R	(χ, D, χ) $\nwarrow$ MPS bond dim. $\nwarrow$ MPO bond dim.
W	(D, d, d, D)

The computationally most expensive part of DMRG is to find the ground state vector Θ of H_{eff} .

Lanczos: Compute $H_{\text{eff}}\Theta$ and iterate

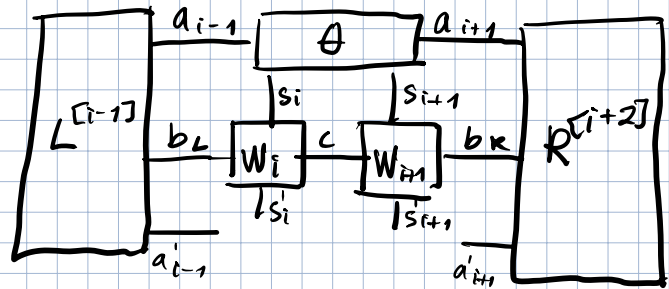
The most costly part is to form H_{eff} .

In general be $\mathcal{O}(\chi^4 d^4)$

With MPS, we can actually compute $H_{\text{eff}}\Theta$ more efficiently without forming the full H_{eff} matrix.

Idea: Choose the order of contractions wisely.

Computing Heff θ



Contract	over	result	cost
1) $L_{a_{i-1} b_L a_{i+1}} \theta_{a_{i-1} s_i s_{i+1} a_{i+1}}$	d_{i-1}	$(a_{i-1}', b_L, s_i, s_{i+1}, a_{i+1})$	$\chi^3 D d^2$
2) $\dots W_{b_L c}^{s_i s_i'}$	b_L, s_i	$(a_{i-1}', s_i', c, s_{i+1}, a_{i+1})$	$\chi^2 D^2 d^3$
3) $\dots W_{c b_R}^{s_{i+1} s_{i+1}'}$	c, s_{i+1}	$(a_{i-1}', s_i', s_{i+1}', b_R, a_{i+1})$	$\chi^2 D^2 d^3$
4) $\dots R_{a_{i+1} b_R a_{i+1}}$	a_{i+1}, b_R	$(a_{i-1}', s_i', s_{i+1}', a_{i+1})$	$\chi^3 D d^2$

= Heff θ

Total cost $O(\chi^3 D d^2 + \chi^2 D^2 d^3)$

$\Rightarrow$ much cheaper compared to $O(\chi^4 d^4)$

(given that $D < \chi d^2$)

($D = 5$ for Heisenberg model, typically small)

The DMRG algorithm (two-site)

- (1) Form θ by contracting the two tensors at the CO over their shared bond.
(The shared bond then disappears, and the entanglement between the two sites can grow up to χd (Schmidt-rank))
- (2) Solve $H_{\text{eff}} \theta = E \theta$ by Lanczos, starting from the current θ .
- (3) Reshape the optimised θ into a $(a_{i-1} s_i) \times (a_{i+1} s_{i+1})$ $(d\chi) \times (d\chi)$ and do an SVD

$$\theta = U \Sigma V^+$$

- (4) Truncate: Keep the χ largest singular values of Σ , record the discarded weight

$$\epsilon_\chi = \sum_{k > \chi} \sigma_k^2$$

and renormalise the remaining singular values.

(5) Move the CO

Sweep to the right: CO goes from $(i, i+1)$ to $(i+1, i+2)$

Site i goes into $L^{[i]}$: $A^{S_i \times n} = U$

New CO: $\theta = \sum V^+ B^{[i+2]}$

Right environment shrinks $R^{[i+2]} \rightarrow R^{[i+3]}$

Similarly, when sweeping to the left:

CO goes to $(i-1, i)$

Site $i+1$ goes into $R^{[i+1]}$: $B^{[i+1]} = V^+$

CO: $\theta = A^{[i-1]} U \Sigma$

Typically:

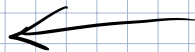
* Start in some right-canonical form

* Form CO on bond $(1, 2)$: $\theta = B^{[1]} B^{[2]}$

* Optimise θ

* Move to the right

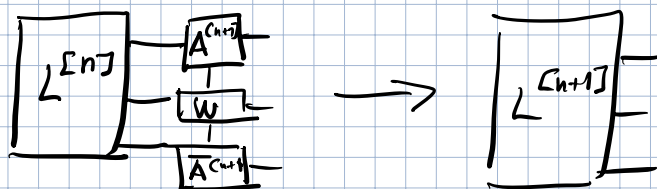
	1	2	3	4	5	6
	R	R	R	R	R	R
Optimize	$\boxed{\theta}$		R	R	R	R
	L	$\boxed{\theta}$		R	R	R
	L	L	$\boxed{\theta}$		R	R
	L	L	L	$\boxed{\theta}$		R
	L	L	L	L	$\boxed{\theta}$	
	L	L	L	$\boxed{\theta}$		R



* Keep sweeping back and forth until the energy converges!

Initialise the environments

* Recursively from left for L
right R



When we construct the environments, we should keep them in memory.

They can be re-used in the backward part of the sweep.

An update on $CO(i, i+1)$ changes $R^{[k]}$ for $k \leq i+1$ and $L^{[k]}$ for $k \geq i$.

Why two-site?

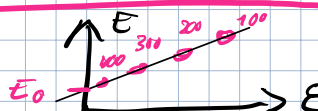
Main reason: One-site update can not increase the bond dimension, so if you start in some state with χ , you stay there.

CO is $\chi \times \chi$ for 1-site

$d\chi \times d\chi$ for 2-site.

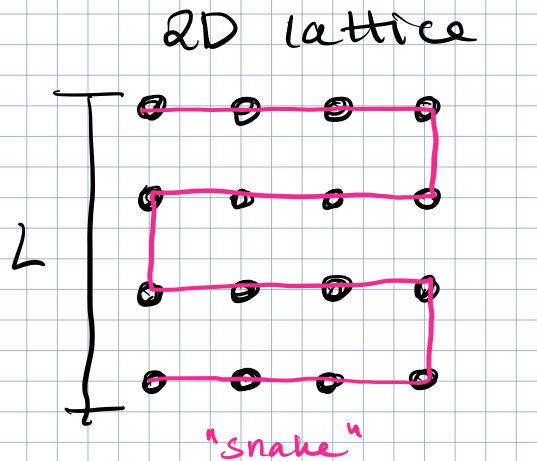
Practicalities

- * Run for several chosen maximal χ (one run per χ)
- * Plot the converged $E(\chi)$ vs E_χ
- * Extrapolate linearly to $E_\chi \rightarrow 0$.



The MPS describes a chain.

For other physical systems, you can map them onto a chain.



(induce longer-range interactions)
(and entanglement entropy $S_A \sim L$)

Chemical molecules
or orbitals:

* You need to choose
an order of the
orbitals, that
gives the "chain".